

AMENDMENT NO. 1 MARCH 1985
TO
IS : 110000 (Part IV)-1980 METHODS OF TESTS FOR
INTERNAL COMBUSTION ENGINES

PART IV DECLARATION OF POWER, EFFICIENCY,
FUEL CONSUMPTION AND LUBRICATING
OIL CONSUMPTION

(Page 3, clause 3.1) — Substitute the following for the existing clause:

' 3.1 A tolerance of +5 percent in fuel consumption at 100 percent of the rated load shall be allowed.'

(Page 3, clause 3.2.3 and Note under it) — Substitute the following for the existing:

'3.2.3 *Adjustment for calorific value of fuel for liquid fuel engines* — In case the liquid fuel engines use fuel different from the fuel with a lower calorific value of 42 000 kJ/kg, a correction for the difference in calorific value given by the following formula shall be applied:

$$\beta_{cv} = \frac{C V}{42\,000} \text{ and}$$

Adjusted specific fuel consumption

$$= \beta_{cv} \times \text{calculated specific fuel consumption}$$

where

β_{cv} = specific fuel consumption adjustment factor for calorific value of fuel; and

$C V$ = net lower calorific value of fuel used, kJ/kg.

Note — The specific fuel consumption of a liquid fuel (diesel) engine is related to a reference lower calorific value of 42 000 kJ/kg. For other fuels the calorific value shall be determined in accordance with 3.2.3.1.

(Page 3, clause 3.2.3) — Add the following new clause after 3.2.3:

' 3.2.3.1 The calorific value of fuel for liquid fuels may be determined in accordance with IS : 1448 [P : 6]-1960 'Methods of tests for petroleum and its products: Part 6 Calorific value by bomb calorimeter method'. Alternatively it may be calculated as per IS : 1448 [P : 7]-1960 'Methods of tests for petroleum and its products: Part 7 Calorific value by calculation.'

(Page 4, clause 6, second sentence) — Substitute the following sentence for the existing:

' Alternatively, it may be calculated by using the indicated power from the indicator diagram or by using Willan's line method. '

(Page 4, clause 7.3.1, Title) — Substitute 'Power adjustment factor 'α' for 'Power correction factor 'α'.

(Page 4, clause 7.3.1.1) Substitute the following formula for the existing:

$$\alpha = \left(\frac{p}{100} \right)^{0.65} \times \left(\frac{300}{T} \right)^{0.5}$$

(Page 5, clause 7.3.1.1.1) — Substitute the formula ' $\alpha = 1 - \frac{A}{100}$ '

for ' $\alpha = 1 + \frac{A}{100}$ '.

(Page 5, clause 7.3.1.2) — Substitute the following formula for the existing:

$$\alpha = \left(\frac{p}{100} \right) \left(\frac{300}{T} \right)^{0.5}$$

(Page 5, clause 8.1) — Substitute the following for the existing clause:

'8.1 A tolerance of + 5 percent in fuel consumption at 100 percent of the rated load throughout the speed range shall be allowed.'

(Page 5, clause 8.2.2 and Note under it) — Substitute the following for the existing:

'8.2.2 Adjustment for calorific value for liquid fuel — It shall be applied when there is a difference in the net calorific value of the test fuel and that of fuel specified in Section II of IS : 10000 (Part 2)-1980. The factor is given by the following formula:

$$\beta_{cv} = \frac{CV}{42\,000}$$

where

β_{cv} and CV have the same connotation as in 3.2.3.

Note — The specific fuel consumption of a liquid fuel (diesel) engine is related to a reference lower calorific value of 42 000 kJ/kg. For other fuels the calorific value shall be determined in accordance with 3.2.3.1.'

(EDC 14)